Infant and Child Foods

Category 1 · Ages 0–5 · 2026 Edition

Government-limit default — Method v2.0

AUTHOR & PUBLISHER

Karen Pendergrass · ORCID 0000-0002-2348-7259
Institute of Contaminant Standards (ICS) · Heavy Metal Tested & Certified

EDITION

2026 web edition · 2026-08-06 · nine subcategories · eight analytes (Pb, iAs, Hg, Cd, Cr, Ni, Sn, Al)

About this manual

The complete Heavy Metal Tested & Certified standard for Category 1, Infant and Child Foods. It presents all nine subcategory standards as one master limit matrix and gives each its full derivation, government by government. Every number traces to a named sovereign limit; nothing is invented. This is the current web edition rendered as a citable, self-contained volume — the numbers are identical to heavymetalcertified.com/manuals/infant-and-child-foods.

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Method v2.0 — how every number is set

Every limit in this manual is the strictest maximum level (ML) set by any credible government regulator worldwide — the European Union, the United Kingdom, Codex Alimentarius, Health Canada, China, India, Brazil, the Gulf states (GSO), Australia/New Zealand, the US FDA, and others — expressed in the product's own native basis. It is a lookup of sovereign law, not a modelled or negotiated figure: a certified product sits at or below every applicable government limit at once.

The powder-to-liquid conversion is applied at the single 8:1 ratio a government actually legislates (China GB 2762), and only within infant formula, where powder and ready-to-feed forms are the same product at two dilutions. Every other subcategory takes government limits at face value in its own basis; no reconstitution factor is invented for a food that is eaten as sold.

The eight analytes are tiered. Tier 1 — lead, cadmium, inorganic arsenic and mercury — are zero-tolerance toxics with no established safe threshold and carry no transitional allowance at any margin. Tier 2 — nickel, aluminium, tin and total chromium — carry the program's 150% transitional allowance under the Status A–E framework, with a stated end date and corrective action.

Chromium is screened as total chromium at the limit shown; hexavalent chromium, the toxic species, is controlled through it — mandatory Cr(VI) speciation on any total-chromium result above the screening trigger, and any quantifiable Cr(VI) is a hard fail. Where an analyte is not regulated for a product and is not material in that matrix it is marked n/m and controlled by this reflex speciation, not assigned an invented number.

Where no government regulates the exact product for an analyte, the limit is read across from the nearest government value for an analogous product, disclosed as such, so a certified product still sits at or below a real sovereign limit rather than an unbounded gap. Limits tighten after publication through the Standards Ratchet as surveillance warrants; they do not loosen.

Notation used throughout the derivations. [1] basis-converted at the 8:1 powder-to-liquid ratio (China GB 2762). [2] read-across: no government sets this analyte for this exact product; the value is read across from the nearest government value for an analogous product. [3] n/m — not material in this matrix; controlled by reflex Cr(VI) speciation, not a standalone number. All values in µg/kg (ppb).

Master limit matrix — all nine subcategories

All values in µg/kg (ppb) in each row's native basis. Split cells show the formulation or matrix variants (cow/soy, rice/non-rice); n/m marks a not-material analyte controlled by reflex speciation. Each subcategory's full derivation follows.

Subcategory	Lead	iAs	Hg	Cd	Cr	Ni	Sn	Al	Basis
Infant formula (powder)	20	20	15	10	2000	250 cow / 400 soy	5000	4000 cow / 8000 soy	as sold
Infant formula (ready-to-feed, liquid)	10	10	4	5	300	100	5000	500 cow / 1000 soy	as consumed
Baby cereals & dry grain products	20	100 rice-based / 20 non-rice	8	40	n/m	3000	5000	500	as sold
Fruit purées	10	20	8	40	n/m	500	5000	500	as consumed
Non-root vegetable purées	10	20	8	40	n/m	500	5000	500	as consumed
Root-vegetable purées	20	20	8	40	n/m	500	5000	500	as consumed
Mixed meals (meat & grain combinations)	10	20	8	40	n/m	500	5000	500	as consumed
Fruit juice (not canned)	20	10	10	10	n/m	250	5000	500	as consumed
Teething biscuits & finger snacks	20	100 rice-based / 20 non-rice	8	40	n/m	3000	5000	500	as sold

Per-subcategory derivations

1 | Infant formula (powder) (as-sold)

Every government maximum in force worldwide, in the as-sold basis; the HMTc limit is the strictest and the Binds column names the law that sets it. A blank cell means that government sets no limit for this analyte in this product.

Analyte	EU	UK	Codex	Canada	China	India	Brazil	GSO	AU/NZ	HMTc	Binds
Lead (Pb)	20	50	80[1]	80[1]	80	200	80[1]	80[1]	80[1]	**20**	EU
Inorganic arsenic (iAs)	20	–	–	–	–	–	160[1]	–	–	**20**	EU
Mercury (tHg)	–	–	–	–	–	–	–	–	–	**15**	IL
Cadmium (Cd)	10	10	–	–	–	100	80[1]	10	–	**10**	EU
Total chromium (Cr)	–	–	–	–	2000	–	–	–	–	**2000**	China
Nickel (Ni)	250	–	–	–	–	–	–	–	–	**250**	EU
Tin (Sn)	–	–	–	–	–	5000	–	–	–	**5000**	India
Aluminium (Al)	–	–	–	–	–	–	–	–	4000[1]	**4000**	AU/NZ

Variants on this page: cow and soy. Where a cell differs between them the master matrix shows both; the derivation below is for cow.

2 | Infant formula (ready-to-feed, liquid) (as-consumed)

Every government maximum in force worldwide, in the as-consumed basis; the HMTc limit is the strictest and the Binds column names the law that sets it. A blank cell means that government sets no limit for this analyte in this product.

Analyte	EU	UK	Codex	Canada	China	India	Brazil	GSO	AU/NZ	HMTc	Binds
Lead (Pb)	10	10	10	10	10	20	10	10	10	**10**	EU
Inorganic arsenic (iAs)	10	–	–	–	–	–	20	–	–	**10**	EU
Mercury (tHg)	–	–	–	–	–	–	–	–	–	**4**	IL
Cadmium (Cd)	5	5	–	–	–	100[1]	10	5	–	**5**	EU
Total chromium (Cr)	–	–	–	–	–	–	–	–	–	**300[2]**	read-across
Nickel (Ni)	100	–	–	–	–	–	–	–	–	**100**	EU
Tin (Sn)	–	–	–	–	–	5000[1]	–	–	–	**5000**	India
Aluminium (Al)	–	–	–	–	–	–	–	–	500	**500**	AU/NZ

Variants on this page: cow and soy. Where a cell differs between them the master matrix shows both; the derivation below is for cow.

3 | Baby cereals & dry grain products (as-sold)

Every government maximum in force worldwide, in the as-sold basis; the HMTc limit is the strictest and the Binds column names the law that sets it. A blank cell means that government sets no limit for this analyte in this product.

Analyte	EU	UK	Codex	Canada	China	India	Brazil	GSO	FDA	HMTc	Binds
Lead (Pb)	20[1]	50[1]	20[1]	–	–	200[1]	20[1]	–	20[1]	**20**	EU
Inorganic arsenic (iAs)	–	–	–	100[1]	200[1]	–	150[1]	–	100[1]	**100**	FDA
Mercury (tHg)	–	–	–	–	–	–	–	–	–	**8[2]**	read-across
Cadmium (Cd)	40[1]	40[1]	–	–	60[1]	100[1]	50[1]	40[1]	–	**40**	EU
Total chromium (Cr)	–	–	–	–	–	–	–	–	–	**n/m[3]**	controlled
Nickel (Ni)	3000[1]	–	–	–	–	–	–	–	–	**3000**	EU
Tin (Sn)	–	–	–	–	–	5000[1]	–	–	–	**5000**	India
Aluminium (Al)	–	–	–	–	–	–	–	–	–	**500[2]**	read-across

Variants on this page: rice-based and non-rice. Where a cell differs between them the master matrix shows both; the derivation below is for rice-based.

4 | Fruit purées (as-consumed)

Every government maximum in force worldwide, in the as-consumed basis; the HMTc limit is the strictest and the Binds column names the law that sets it. A blank cell means that government sets no limit for this analyte in this product.

Analyte	EU	UK	China	India	Brazil	GSO	FDA	HMTc	Binds
Lead (Pb)	10000[1]	50[1]	–	100[1]	20[1]	100[1]	10[1]	**10**	FDA
Inorganic arsenic (iAs)	20[1]	–	200[1]	–	150[1]	–	–	**20**	EU
Mercury (tHg)	–	–	–	–	–	–	–	**8[2]**	read-across
Cadmium (Cd)	40[1]	40[1]	60[1]	–	100[1]	40[1]	–	**40**	EU
Total chromium (Cr)	–	–	–	–	–	–	–	**n/m[3]**	controlled
Nickel (Ni)	500[1]	–	–	–	–	–	–	**500**	EU
Tin (Sn)	–	–	–	–	–	–	–	**5000[2]**	read-across
Aluminium (Al)	–	–	–	–	–	–	–	**500[2]**	read-across

5 | Non-root vegetable purées (as-consumed)

Every government maximum in force worldwide, in the as-consumed basis; the HMTc limit is the strictest and the Binds column names the law that sets it. A blank cell means that government sets no limit for this analyte in this product.

Analyte	EU	UK	India	GSO	FDA	HMTc	Binds
Lead (Pb)	20[1]	50[1]	200[1]	—	10[1]	**10**	FDA
Inorganic arsenic (iAs)	20[1]	—	—	—	—	**20**	EU
Mercury (tHg)	—	—	—	—	—	**8**[2]	read-across
Cadmium (Cd)	40[1]	40[1]	100[1]	40[1]	—	**40**	EU
Total chromium (Cr)	—	—	—	—	—	**n/m**[3]	controlled
Nickel (Ni)	500[1]	—	—	—	—	**500**	EU
Tin (Sn)	—	—	5000[1]	—	—	**5000**	India
Aluminium (Al)	—	—	—	—	—	**500**[2]	read-across

6 | Root-vegetable purées (as-consumed)

Every government maximum in force worldwide, in the as-consumed basis; the HMTc limit is the strictest and the Binds column names the law that sets it. A blank cell means that government sets no limit for this analyte in this product.

Analyte	EU	UK	India	GSO	AU/NZ	FDA	HMTc	Binds
Lead (Pb)	20[1]	—	100[1]	100[1]	—	20[1]	**20**	EU
Inorganic arsenic (iAs)	20[1]	—	—	—	—	—	**20**	EU
Mercury (tHg)	—	—	—	—	—	—	**8**[2]	read-across
Cadmium (Cd)	40[1]	40[1]	—	40[1]	100[1]	—	**40**	EU
Total chromium (Cr)	—	—	—	—	—	—	**n/m**[3]	controlled
Nickel (Ni)	500[1]	—	—	—	—	—	**500**	EU
Tin (Sn)	—	—	—	—	—	—	**5000**[2]	read-across
Aluminium (Al)	—	—	—	—	—	—	**500**[2]	read-across

7 | Mixed meals (meat & grain combinations) (as-consumed)

Every government maximum in force worldwide, in the as-consumed basis; the HMTc limit is the strictest and the Binds column names the law that sets it. A blank cell means that government sets no limit for this analyte in this product.

Analyte	EU	UK	Codex	China	India	Brazil	GSO	FDA	HMTc	Binds
Lead (Pb)	10000[1]	—	20[1]	—	200[1]	20[1]	—	10[1]	**10**	FDA
Inorganic arsenic (iAs)	20[1]	—	—	200[1]	—	150[1]	—	—	**20**	EU
Mercury (tHg)	—	—	—	—	—	—	—	—	**8**	IL
Cadmium (Cd)	40[1]	40[1]	—	60[1]	100[1]	50[1]	40[1]	—	**40**	EU
Total chromium (Cr)	—	—	—	—	—	—	—	—	**n/m³**	controlled
Nickel (Ni)	500[1]	—	—	—	—	—	—	—	**500**	EU
Tin (Sn)	—	—	—	—	5000[1]	—	—	—	**5000**	India
Aluminium (Al)	—	—	—	—	—	—	—	—	**500[2]**	read-across

8 | Fruit juice (not canned) (as-consumed)

Every government maximum in force worldwide, in the as-consumed basis; the HMTc limit is the strictest and the Binds column names the law that sets it. A blank cell means that government sets no limit for this analyte in this product.

Analyte	EU	UK	Codex	Canada	India	GSO	FDA	HMTc	Binds
Lead (Pb)	20[1]	30[1]	30[1]	—	50[1]	30[1]	50[1]	**20**	EU
Inorganic arsenic (iAs)	20[1]	—	—	30[1]	—	—	10[1]	**10**	FDA
Mercury (tHg)	—	—	—	—	—	—	—	**10**	IL
Cadmium (Cd)	20[1]	—	—	—	—	—	—	**10**	IL
Total chromium (Cr)	—	—	—	—	—	—	—	**n/m³**	controlled
Nickel (Ni)	250[1]	—	—	—	—	—	—	**250**	EU
Tin (Sn)	—	—	—	—	—	—	—	**5000[2]**	read-across
Aluminium (Al)	—	—	—	—	—	—	—	**500[2]**	read-across

9 | Teething biscuits & finger snacks (as-sold)

Every government maximum in force worldwide, in the as-sold basis; the HMTc limit is the strictest and the Binds column names the law that sets it. A blank cell means that government sets no limit for this analyte in this product.

Analyte	EU	UK	Codex	Canada	China	India	Brazil	GSO	AU/NZ	FDA	HMTc	Binds
Lead (Pb)	20[1]	50[1]	20[1]	–	–	200[1]	20[1]	–	–	20[1]	**20**	EU
Inorganic arsenic (iAs)	150[1]	–	–	100[1]	200[1]	–	150[1]	100[1]	–	100[1]	**100**	FDA
Mercury (tHg)	–	–	–	–	–	–	–	–	–	–	**8[2]**	read-across
Cadmium (Cd)	40[1]	40[1]	–	–	60[1]	100[1]	50[1]	40[1]	100[1]	–	**40**	EU
Total chromium (Cr)	–	–	–	–	–	–	–	–	–	–	**n/m³**	controlled
Nickel (Ni)	3000[1]	–	–	–	–	–	–	–	–	–	**3000**	EU
Tin (Sn)	–	–	–	–	–	5000[1]	–	–	–	–	**5000**	India
Aluminium (Al)	–	–	–	–	–	–	–	–	–	–	**500[2]**	read-across

Variants on this page: rice-based and non-rice. Where a cell differs between them the master matrix shows both; the derivation below is for rice-based.

Edition record

The dated editions of this manual. Each is kept at a stable permalink so a regulator or brand can reconstruct exactly what the manual required on any date. This 2026 web edition is the current standard set; the 384-page deposited editions are prior, superseded volumes and are not served against it.

Edition	Date	Status	DOI	What changed
2026 web edition	2026-08-06	current	pending (Crossref)	Reorganized to the nine rendered subcategories under the government-limit default (Method v2.0). Consolidated, version-pinned PDF now published (v2); a DOI is minted on deposit.
2026 deposit (rev.)	2026-06-20	superseded	10.5281/zenodo.20771543	384-page deposited volume; revision incorporating the Expedited Baseline (Governance §4.5) and Lot Testing Schedule v2.1.
2026 deposit (v0)	2026-05-18	superseded	10.5281/zenodo.20270512	First public deposit of the standalone 384-page volume.

The concept DOI 10.5281/zenodo.20270511 always resolves to the latest deposited edition. The current 2026 web edition has no DOI yet (Crossref deposit pending); this manual is citable by its permalink in the meantime.

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